The Effect of CME Gold Futures and ICE Dollar Index Futures on the KRX Gold Market

Myeong-Hoon Yeom

Retail Strategy Team, Kiwoom Securities Co.

Jae-Seung Baek *

Department of International Finance, Hankuk University of Foreign Studies

Abstract

The KRX (Korea Exchange) gold market opened in March 2014 according to the government policy legalizing financial transactions, and traded one-gram unit of real gold by Korean currency (KRW) in the exchange market. Despite the fact that the KRX gold market shows high efficiency in terms of taxes and fees in contrast to the existing gold market, studies on the KRX gold market have been rarely conducted until quite recently. We explore the KRX gold market price efficiency by showing price discovery factors in the market. According to the existing studies, the futures market has a characteristic of leading the spot market. Therefore, this study put forward a hypothesis that the return rate of the gold in the CME (Chicago Mercantile Exchange), in which gold futures are traded at night in Korea Standard Time, would have a positive impact on the return of the KRX gold market, which starts at 9:00am the following day. In addition, this paper goes into detail about the international gold price having a negative correlation with the value of the United States (US) dollar. To test the goal of the research, both the return rate of CME gold futures at t-1 days and ICE (Intercontinental Exchange) dollar index futures at t-1 days are investigated to determine whether they have an impact on the market rate of return in the KRX gold market at t day using regression analysis and vector auto-regression (VAR) model in order to look closely at the factors that determine the market price of the KRX gold market. First, the return rate of CME gold futures at t-1 days has a positive impact of significance on the market rate of return in the KRX gold market at t day. Second, the return rate of ICE dollar index futures at t-1 days could not be confirmed to significantly determine the market rate of return in the KRX gold market at t day.

Key Words KRX Gold Market; CME Gold Futures; ICE Dollar Index Futures; VAR Model

JEL classification: G10, G14, G15

*Corresponding author: Jae-Seung Baek, Department of International Finance, Hankuk University of Foreign Studies, 81, Oedae-ro, Mohyun, Yongin-si, Gyeonggi-do, 17035, Korea, Tel: +82-31-330-4936, Fax: +82-31-330-4074, email: jbaek@hufs.ac.kr.

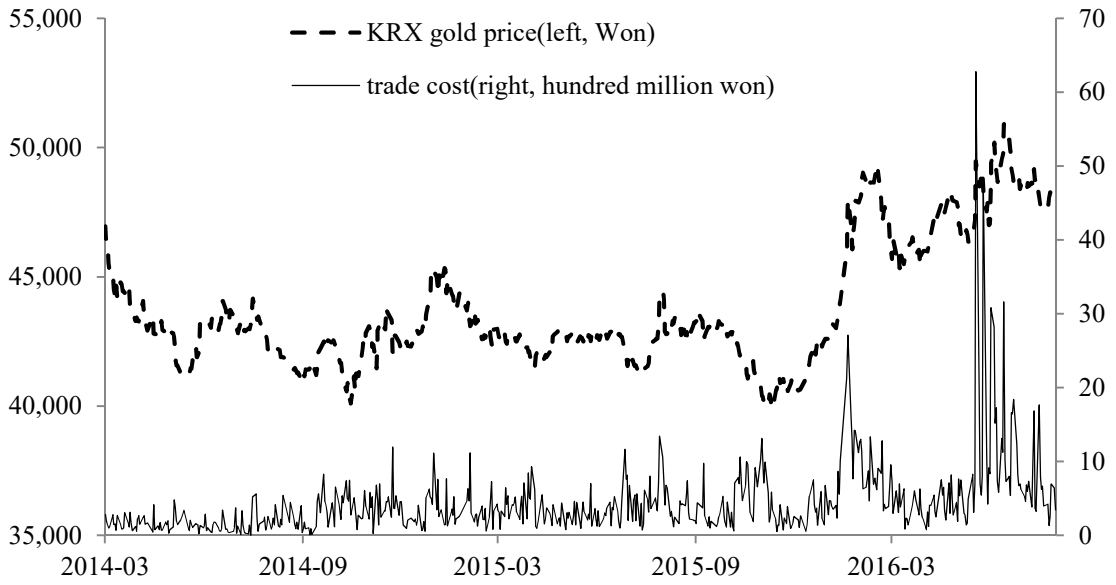
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1. Introduction

With the element symbol Au and atomic number 79, gold is not only the investment asset which investors prefer most but also real currency regardless of ages and countries. gold is a material with a beautiful color which does not corrode and is easily processed and so gold is used in many fields including jewelry, real currency, industry, medical treatment, etc. For a long time, men have wished to possess gold because of these characteristics.

Kim et al. (2015) explores various statistical approaches to examine the economic index which predicts gold prices and the variables having an effect on gold prices and compared their results to existing research related to gold. Hong (2013) also investigates factors having an effect on the amount of gold transactions so that the rise of variability in the United States (US) stock market positively affected the demand of gold futures. According to Lee (2014), gold variability has precedence over the variability index. Solt and Swanson (1981), Sjaastad and Scacciavillani (1996), and Xu and Fung (2005) conduct studies on the price determination factors of gold, and posit that there is a relationship between variability, the prime exchange rate, and gold price. As Hiller et al. (2006) points out, in the case of including gold in an investment portfolio as the representative of safe property, gold is said to have a diversified investment effect, lowering variability of the return rate of the portfolio.

Figure 1 KRX gold price and trade cost transition
The left axis is KRX gold price and the unit is won (KRW), while the right axis is the amount of KRX gold market transaction and the unit is one hundred million won. The sample period is from the 24th March, 2014 to the 31st August, 2016.



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The methods the Korean investor could directly invest in gold before the day that the KRX (Korea Exchange) gold market opened on March 2014 were mainly in the wholesale and retail jewelry shops, gold banking in a bank, etc. Although the KRX gold market is implied to be highly efficient in regards to fees and taxes the investor must pay, there is little research supporting this.

In this research, the KRX gold market is examined over 2 years and CME (Chicago Mercantile Exchange) gold futures and ICE (Intercontinental Exchange) dollar index futures are also considered in terms of their influence on the KRX gold market. As shown in Figure 1, the amount of a KRX gold market transaction appeared to be an average of 24 million won (Korean currency, Korea Won, *KRW*) a day in the opening year (2014). It grew constantly thereafter and became more than three times bigger than in the opening year, with the average trade cost being 77 million won (KRW) a day in 2016.

Barclay and Hendershott (2003) and Yeom et al. (2013) verify the price discovery effect by analyzing the significance of the influence between spot and futures time difference. As stated in Chan (1992), Lee and Lim (2002), Tse et al. (2006), and Yeom and Baek (2015), it is commonly accepted that the futures exchange market is leading the spot market. This study investigates the significance of influence between the variables and time difference. To test the hypotheses of the paper, we investigate whether the return rate of gold derivatives in the United States at t-1 days significantly affects the gold spot market in Korea at t day. The return rate of CME gold futures at t-1 days is analyzed to determine whether there is a price discovery effect on the market rate of return of the KRX gold market by using the daily data from the 24th March, 2014 to the 23rd August, 2014 for the empirical analysis. Moreover, in order to verify the proposition that the international price of gold and the value of the US dollar have a negative correlation, this study preforms an empirical analysis measuring the influence that the return rate of the ICE dollar index futures at t-1 days had on the market rate of return of the KRX gold market at t day.

The empirical analysis result utilizing the regression analysis and VAR model represents that the rate of return of CME gold futures at t-1 days positively affects the market rate of return of the KRX gold market at t day, while the significance of the return rate of the ICE dollar index futures influencing the market rate of return of the KRX gold market was not clear. This result shows that the rate of return of the CME gold futures has the leading market rate of return of the KRX gold market, and the KRX gold market will be able to be hedged by utilizing CME gold futures. Conversely, our study develops the general concept that the US dollar value is operated contrary to the gold price, and examines the influence of the variable with time difference, that

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is, whether the return rates of ICE dollar index futures at t-1 days influence on the market rate of return of the KRX gold market at t day, but the statistical significance was unable to be found.

The rest of the paper is organized as follows. Section 2 introduces the KRX gold market briefly. Section 3 describes research data and method to test the hypotheses. Section 4 explains the results and section 6 outlines the conclusions of the study.

2. Overview of KRX Gold Market

2.1 Trading System of KRX Gold Market

According to the government policy for the legalization of gold transaction and the development of the precious metal industry, the KRX gold market utilized the HTS (home trading system) for the opening of the gold spot market making it possible to make real-time transactions, similar to stock trading. The role and goals of the KRX gold market were realized as the legalization of the underground economy through the eradication of illegal transactions, reliability and enhancement of price and quality, and preparation for the developing foundations of the precious metal industry were announced by the Financial Services Commission, the Ministry of Strategy and Finance, the Ministry of Trade, Industry and Energy, National Tax Service, and the Office of Customs Administration in July, 2013.

In the KRX gold market, there are a number of gold traders such as importers of gold, gold refiners, normal investor, etc. The traders deposit gold in the Korea Securities Depository, which is the custody facility. Meanwhile, gold buyers such as workmanship traders, the industry, normal investors, etc. have to deposit the cash being used to purchase the gold into a KRX gold account which is set up by the securities company. A normal investor with non-membership sends an order into the exchange trading system through the securities company, who is the general member of the KRX gold market, for trading in the KRX gold market.

2.2 Trading System in KRX Gold Market

Gold bullion, which is the sales object of the KRX gold market, has 99.99% purity and its weight, as a raw material in gold bar form, is one kilogram. The eligible manufacturer and importer whom the Korea Exchange chooses to supply, has to pass a quality standard which the Korea Exchange and quality assurance agency choose. On the surface of a gold bullion are engraved the name of the refining industry, a trademark, purity, weight, manufacture’s serial number, etc.

The unit of transaction is one gram and the asking price is 10 won (KRW). The price is shown as won per 1g in the KRX gold market. A limit order is feasible and the maximum quantity per one bid is 5kg. The price limits are expressed as an upper price limit and a lower

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price limit. The standard price determines the restriction of the price range expressed as the upper limit price and lower price limit. The standard price is based on the latest closing price from the day before, just before the beginning of a transaction day. The daily restriction of the price range is ±10% of the standard price. The trading hours of the KRX gold market are from 9:00am until 15:30pm. 8:00 am until 9:00 am, that is, one hour before the transaction day starts, is the time frame for making a decision about buying at opening, while from 3:20 pm to 3:30 pm, that is, 10 minutes before closing the transaction day, is the time frame for making a decision about buying at closing.

Regarding the KRX gold market, a transaction is connected through competitive sales, according to the 'price-time' priority between several asking prices, similar to the stock market. It is concluded that an execution of sales deal is feasible as soon as the asking price comes in. However, the trading methods of buying on opening and closing prices are accomplished using the singular price acquisition method. The payment in the KRX gold market is a same day settlement (t day) and the deposit and withdrawal of the gold bullion is feasible in 1kg units. If the withdrawal of the gold is requested of the custody facility through the trading securities company, the custody facility pays the gold to the fixer.

Table 1 Trading System in the KRX Gold Market

	Contents
Type	Purity: 99.99%, Weight: 1kg Gold Bullion
Trade Unit	1g (Asking Unit: 1g)
Trading Hours	09:00am ~ 15:30pm
Price Mark	Won (KRW) per 1g
Asking Price Unit	10 Won (KRW)
Price Limits	Closing Price of the Previous Day ±10%
Type of Order	Limited Order
Order Limit	5,000g (5kg) per bid
Settlement Day	Same Day Settlement (t day)
Customer Margin	Cash Margin 100%

2.3 Characteristics of KRX Gold Market

The KRX gold market can invest in gold with a lower cost compared with the existing gold market, and can utilize HTS on a real time basis, similarly to stock trading, making it possible to deal with small (1g) gold transactions. There are many advantages to the KRX gold market. First, the KRX gold market has the advantage in terms of price when taking into consideration transaction costs including fees, taxes, purchase spread, etc. compared with the existing gold over-the-counter market (outside wholesale markets, the bank gold banking, jewelry shops,

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etc.). There is a tax benefit in which the transfer income tax is not imposed and is not included in the object of comprehensive taxation on financial income. As to the gold-related fund and gold banking, a 15.4% of profit income tax is imposed.

Second, if the spot gold is not delivered from a warehouse, the value added tax is not imposed at the KRX gold market. Therefore, in the case of using the KRX gold market for the investment purpose of forecasting the gold price rise, the investment efficiency, in comparison with the other gold markets, is high. As shown in Table 2, when it is assumed that the price rises 10% shortly after purchasing the rate of return on investment using the KRX gold market is higher than the other markets. That is, the rate of return in the KRX gold market is 9.6%, the account transaction of gold banking is 6.6%, the spot transaction of gold banking -5.6%, and the rate of return of the jewelry shop outside the wholesale market is -2.2%.

Table 2 Comparison of return rates according to the gold market types
The closing price of the KRX gold market represents the purchasing price of gold, final purchase price of the Shinhan Bank, and the officially fixed price of the Korea Federation of Jewelry Sales Business by standards on the 10th April of 2015. The unit is by won (KRW).

	Purchasing Price	Buyout Fees	Sale Price	Sale Commission	Income Tax	Surtax	Rate of Return (%)
KRX Gold Market	42,500	128	46,750	140			9.6
Gold Banking (Account Transaction)	42,742		46,085		515		6.6
Gold Banking (Spot Transaction)	44,435		46,398			4,444	-5.6
Outside Whole Sale	43,040		46,398			4,304	-2.2

3. Data and methodology

3.1 Research Data

Where there is time difference between the variables, Granger (1969, 1980) interprets this kind of dataset as a lead-lag relationship in case there is a cause-and-effect relationship existing in the situation. Moreover, considering the existing research that suggests that the futures market leads the spot market, it can be estimated that the return rate of CME gold futures and of ICE dollar index futures traded at night Korean time have an effect on the market rate of return of the KRX gold market when it opens at 09:00am the following day. That is, the prices

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of CME gold futures and of ICE dollar index futures traded when it is night time in Korea can have an effect on the market rate of return of the KRX gold market when it opens because the price reflects the financial market news of the United States and major European countries. In Table 3, the business hours of CME gold futures, ICE dollar index futures, and the KRX gold market are arranged.

Table 3 Description of the Variables
Daylight Savings Time in the United States is applied from the second Sunday in March until 2:00am on the first Sunday of November. Business hours are based on Korean time.

Panel A: Business Hours of Markets

Division	Daylight Saving Time Off	Daylight Saving Time
KRX Gold Market	09:00am ~ 15:30pm	The same
CME Gold Futures	08:00am ~ the following day 07:00am	07:00am ~ the following day 06:00am
ICE Dollar Index Futures	10:00am ~ the following day 07:00am	09:00am ~ the following day 06:00am

Panel B: Variable Descriptions

Variable	Description
$KRX_{co,t}$	Rate of Return of KRX at t day (= KRX Gold Market Opening price at the t day/Closing Price at t-1 days)
$CME_{cc,t-1}$	Rate of Return of CME Gold Futures at t-1 days (= CME Gold Futures Closing Price at t-1 days/Closing Price at t-2 days)
$ICE_{cc,t-1}$	Return Rate of ICE Dollar Index Futures at t-1 days (= ICE Dollar Index Futures Closing Price at t-1 days/Closing Price at t-2 days)

In order to utilize the regression analysis and VAR model to provide empirical evidence, the daily data of the market rates of return in the KRX gold market from the 24th March 2014 to the 23rd August 2016, is set to the dependent variable. The market rate of return in the KRX gold market equals the opening price at t day divided by the closing price at t-1 days minus 1. Moreover, after setting the data of daily return rates of both the CME gold futures and ICE dollar index futures to the independent variable, it is examined whether the independent variables have any significant effects on the market rates of returns of the KRX gold market, which are the dependent variable. The rate of return of CME gold futures at t-1 days and the return rate of ICE dollar index futures at t-1 days are included by calculating the closing price at t-1 days after dividing the t-1 days closing price into the closing price at t-2s day minus 1.

Table 4 shows the summary statistics of the market rates of return of the KRX gold market at t day, the return rates of CME gold futures at t-1 days, and the return rates of ICE dollar index futures at t-1 days. The absolute value of the maximum and minimum values of CME gold futures and the KRX gold market, which are based on gold price, are greater than ICE dollar index futures because the variability of gold price is greater than the dollar variability.

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Table 4 Summary statistics

Sampling period is from the 24th March, 2014 to the 23rd August, 2016. $KRX_{co,t}$ is the market rate of return of the KRX gold market at t day, $CME_{cc,t-1}$ is the rate of return of CME gold futures at t-1 days, $ICE_{cc,t-1}$ is the rate of return of ICE dollar index futures at t-1 days. S

Division	Minimum	Maximum	Mean	Standard Deviation	Variance
$KRX_{co,t}$	-6.561	2.990	0.134	0.679	0.461
$CME_{cc,t-1}$	-3.140	6.211	0.006	0.978	0.956
$ICE_{cc,t-1}$	-2.222	2.460	0.029	0.529	0.280

Table 5 represents the result of the interrelation between three variables: the market rate of return of the KRX gold market at t day, the return rate of CME gold futures at t-1 days, and the rate of return of ICE dollar index futures at t-1 days. The analysis result of the interrelation is that the market rate of return of the KRX gold market ($KRX_{co,t}$) and the return rate of CME gold futures at t-1 days ($CME_{cc,t-1}$) positively relate at 0.576 with a 1% level of significance. The correlation between the market rate of return of the KRX gold market at t day ($KRX_{co,t}$) and the return rate of ICE dollar index futures at t-1 days ($ICE_{cc,t-1}$) was negative (-0.147) with significance at the 1% level. It is noticeable that the negative correlation between the return rate of CME gold futures at t-1 days and ICE dollar index futures at t-1 days (-0.344) is more negative than the correlation between the market rate of return of the KRX gold market at t day ($KRX_{co,t}$) and the return rate of CME gold futures at t-1 day ($ICE_{cc,t-1}$; -0.147) because the business hours of CME gold futures and ICE dollar index futures almost coincided with each other and night time in Korea Standard Time.

Table 5 Correlation Analysis

This table presents Pearson correlations among rate of returns from the 24th March, 2014 to the 23rd August, 2016. $KRX_{co,t}$ is the market rate of return of the KRX gold market at t day, $CME_{cc,t-1}$ is the return rate of CME gold futures at t-1 days, $ICE_{cc,t-1}$ is the return rate of ICE dollar index futures at t-1 days. *** denotes the 1% significance level.

Division	$KRX_{co,t}$	$CME_{cc,t-1}$	$ICE_{cc,t-1}$
$KRX_{co,t}$	1	0.576***	-0.147***
$CME_{cc,t-1}$		1	-0.344***
$ICE_{cc,t-1}$			1

3.2 Research Method

According to the existing research, the futures market can perform the price discovery function and the price leading function over the spot market. Yeom and Baek (2015) found that the return rate of future stock at t-1 days can have a price discovery effect on the spot market at t day in the case that the return rate of the stock price index futures at t-1 days affected the return rate of the spot market at t day

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in Korea. For this purpose, the result of their empirical analysis showed the return rate of KOSPI200 night futures at t-1 days affected the significance on the market rate of the return of KOSPI at t day, and they interpreted that the derivatives which precede time differences had price leading effects or price discovery functions over the spot market which lags in time difference. In addition, related research conducted by Rosenberg and Traub (2009), Chen and Gau (2010), and Choudhary and Bajaj (2013) showed that the futures market performed at a higher price discovery function than the spot market.

This study performs Hypothesis 1 (the return rate of CME gold futures at t-1 days positively influences the market rate of return of the KRX gold market at t day) from the estimation that the return rate of CME gold futures at t-1 days positively influences the market rate of return of the KRX gold market at t day. This estimation is created based on the results of existing research.

Meanwhile, Capie et al. (2005) and Kaul and Sapp (2007) noted that the gold price moved in the opposite direction of the value of the dollar. Park and Han (2014) suggested that the value of the dollar had a negative effect on the international price of gold based on a sampling period where they examined the factors that had an effect on the gold price decision. They determined that the negative correlation between the value of the dollar and the international price of gold appeared more clearly in a specimen period when the international price of gold increased rapidly compared to a more stable specimen period. It was generally reported that the international price of gold negatively correlated with the value of the dollar in news article[1], etc.

In order to verify this estimation, Hypothesis 2 (the return rate of ICE dollar index futures at t-1 days has a negative influence on the market rate of return of the KRX gold market at t day) is formed based on the concept that the return rate of ICE dollar index futures at t-1 days negatively influences the market rate of return of the KRX gold market at t day. Equation (1) is prepared in order to verify Hypothesis 1 and Hypothesis 2. In this research, as shown in equation (1), a regression analysis is utilized in order to analyze the influence that the return rate of CME gold futures and ICE dollar index futures, the t-1 days independent variables, had on the market rate of return of the KRX gold market at t day as the dependent variable. CO means "Close to Open, market rate of return," and CC means "Close to Close, the rate of return".

Hypothesis 1: The return rate of CME gold futures at t-1 days has a positive influence on the market rate of return of the KRX gold market at t day.

[1] Referencing 'International Price of Gold, 0.4% Rising With the Weak Dollar' (Money Today 02/09/2016), 'International Price of Gold, Falling With the Strong Dollar' (News Tomato 31/08/2016), 'International Price of Gold, Collapsing 1,320 dollars With the Strong Dollars' Influence' (Money Today 31/08/2016), etc.

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Hypothesis 2: The return rate of ICE dollar index futures at t-1 days has a negative influence on the market rate of return of the KRX gold market at t day.

$$KRX_{co,t} = \beta_0 + \beta_1 CME_{cc,t-1} + \beta_2 ICE_{cc,t-1} + e_t \quad (1)$$

$$R_t = (KRX_{co,t}, CME_{cc,t}, ICE_{cc,t})^T, \quad R_t = C + \Sigma_{s=1\ldots m} A_s R_{t-s} + e_t \quad (2)$$

Regression equation (1) includes simple key variables in order to provide important information about the effect of the return rate of CME gold futures and ICE dollar index futures on the market rate of return of the KRX gold market. By using the unrestricted VAR model represented in equation (2) in order to verify possible statistic model error, we conducted robustness testing regarding the price discovery function about the market rate of return of the KRX gold market and of the return rates of CME gold futures and ICE dollar index futures. R_t is a 3x1 vector of the three variables of return rate: the market rate of return of the KRX gold market, and the return rates of CME gold futures and ICE dollar index futures while C and A_s are the coefficient matrix of 3x1, 3x3 per each. In addition, m is the length of lag, and e_t is a vector of prediction error.

4. Empirical Results

4.1 Regression Analysis

To perform an empirical test on the daily data of market rates of return of the KRX gold market, the return rates of CME gold futures, and the return rates of ICE dollar index futures, we considered that specific tendencies can exist because of the influence of a business cycle, etc. Therefore, prior to the empirical analysis, as shown in Panel A of Table 6, when the autocorrelation analysis of each variable and the partial correlation analysis are performed, no correlation of estimated residual in the specimen of each variable could be utilized, making results more reliable. The time difference is determined using up to 18 items at maximum, analyzed on the basis of the lowest value of SIC (standard industry classification).

Moreover, as shown in Panel B of Table 6, in order to confirm the probability of time series data, the result of performance of the unit root test (ADF Test; Augmented Dicky-Fuller Test) is interpreted. It is determined that there is no unit root or spurious regression phenomenon in the specimen due to dismissing the hypothesis of the unit root existence on all variables of significance at the 1% level or below.

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Table 6 Autocorrelation Analysis and ADF Test

This table shows the results of autocorrelation analysis and ADF test. Sample period is from the 24th March, 2014 to the 23rd August, 2016. $KRX_{co,t}$ is the market rate of return of the KRX gold market at t day, $CME_{cc,t}$ is the return rate of CME gold futures at t day, $ICE_{cc,t}$ is the return rate of ICE dollar index futures at t day. The time difference is determined using up to 18 items at maximum, analyzed on the basis of the lowest value of SIC.

Panel A: Auto-correlation

	Autocorrelation			Partial Autocorrelation		
구분	$KRX_{co,t}$	$CME_{cc,t}$	$ICE_{cc,t}$	$KRX_{co,t}$	$CME_{cc,t}$	$ICE_{cc,t}$
1	-0.019	-0.054	-0.058	-0.019	-0.054	-0.058
2	0.017	-0.018	0.037	0.016	-0.021	0.033
3	0.053	0.009	-0.059	0.053	0.007	-0.055
4	0.050	0.019	0.011	0.052	0.019	0.003
5	-0.036	0.068	-0.022	-0.035	0.070	-0.018
6	0.064	0.005	-0.085	0.058	0.014	-0.092
7	0.069	-0.017	0.039	0.068	-0.013	0.032
8	0.044	0.001	0.006	0.047	-0.002	0.013
9	0.033	0.021	0.058	0.030	0.017	0.048
10	0.049	-0.014	-0.061	0.036	-0.016	-0.052
11	-0.060	-0.014	0.054	-0.067	-0.016	0.042
12	0.043	-0.061	-0.057	0.033	-0.062	-0.050
13	-0.038	0.027	0.002	-0.048	0.019	-0.007
14	0.035	0.053	-0.060	0.027	0.052	-0.049

Panel B: Unit Root Test

Division	$KRX_{co,t}$	$CME_{cc,t}$	$ICE_{cc,t}$
Time Parameter	0	0	0
t-Statistics	-24.86	-25.68	-25.78
Significance probability	0.00	0.00	0.00

The return rate of CME gold futures and ICE dollar index futures at t-1 days is set to the independent variable and the market rate of return of the KRX gold market is set to the dependent variable in order to examine the influence that CME gold futures and ICE dollar index futures traded in the futures exchange market in the United States, at night time in Korea, has on the KRX gold market in which the transactions don't started until 9:00am on the next business day in Korean time. The regression analysis results are shown in Table 7. As a result of the analysis, the return rate of CME gold futures at t-1 days ($CME_{cc,\ t-1}$) has a positive effect on the market rate of return of the KRX gold market with 1% level of significance. This means that Hypothesis 1 (The return rate of CME gold futures at t-1 days positively affects the market rate of return of the KRX gold market at t day) is established. However, the significance of ICE dollar index futures at t-1 days ($ICE_{cc,\ t-1}$) affecting the market rate of return of the KRX gold market at t day was not clear. Therefore, Hypothesis 2 (The return rate of ICE

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dollar index futures at t-1 days negatively affects the market rate of return of the KRX gold market at t day) could not be confirmed.

Table 7 Regression Analysis

This table represents the effect of CME gold futures and ICE dollar index futures on KRX gold market by regression analysis. Sample period is from the 24th March, 2014 to the 23rd August, 2016. $CME_{cc,t-1}$ is the return rate of CME gold futures at t-1 days, $ICE_{cc,t-1}$ is the return rate of ICE dollar index futures at t-1 days. *** denotes the 1% significance level.

Division	Coefficient	t-Statistics	Significance probability	Tolerance
(Constant)	0.130***	5.712	0.000	
$CME_{cc,t-1}$	0.414***	16.707	0.000	0.882
$ICE_{cc,t-1}$	0.075	1.629	0.100	0.882
Adjusted R^2	0.333	Durbin-Watson		1.930

4.2 Empirical Analysis utilizing VAR Model

The estimated result of the VAR model examining the influence that the return rate of CME gold futures and ICE dollar index futures traded at Korean night time have on the market rate of return of the KRX gold market is shown in Table 8. As a result of this analysis, that examined the significance that the return rate of CME gold futures at t-1 days had on the market rate of return of the KRX gold market at t day in order to verify the Hypothesis 1 (The return rate of CME gold futures at t-1 days positively affects the market rate of return of the KRX gold market at t day), the return rate of CME gold futures at t-1 days reveals that below the 1% level of significance there is a positive correlation (0.414) about the market rate of return of the KRX gold market at t day. Since the return rate of CME gold futures at t-1 days has the price discovery function and the leading effect about the market rate of return of the KRX gold market at t day, this result is interpreted as one that coincided with the result of the regression analysis mentioned above.

In addition, our research examines the influence that the return rate of ICE dollar index futures at t-1 days has on the market rate of return of the KRX gold market at t day using the estimation of the VAR model in order to verify Hypothesis 2 (The return rate of ICE dollar index futures at t-1 days negatively affects the market rate of return of the KRX gold market at t day). The result, just like the result of the regression analysis mentioned above in Table 7, is of low significance. This indicates that there exists no significance relationship between the return rate of ICE dollar index futures and KRX gold market price.

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Table 8 Unrestricted VAR Result

This table presents unrestricted VAR results among all exchanges. Sample period is from the 24th March, 2014 to the 23rd August, 2016. $KRX_{co,t}$ is the market rate of return of the KRX gold market at t day, $CME_{cc,t}$ is the return rate of CME gold futures at t day, $ICE_{cc,t}$ is the return rate of ICE dollar index futures at t day. ***, ** denote the 1% and 5% significance level, respectively.

Division		$KRX_{co,t}$	$CME_{cc,t}$	$ICE_{cc,t}$
$KRX_{co(t-1)}$	Coefficient	0.032	-0.021	0.093**
	Standard Deviation	(0.041)	(0.073)	(0.039)
	t-value	[0.787]	[-0.291]	[2.381]
$CME_{cc(t-1)}$	Coefficient	0.410***	-0.0659	-0.018
	Standard Deviation	(0.025)	(0.0440)	(0.024)
	t-value	[16.405]	[-1.498]	[-0.782]
$ICE_{cc(t-1)}$	Coefficient	0.070	-0.0484	-0.063
	Standard Deviation	(0.046)	(0.0814)	(0.044)
	t-value	[1.509]	[-0.596]	[-1.440]
Adjusted R^2		0.329	0.003	0.006
F- Statistics		49.317	0.712	1.616

This study also performs variance decomposition and impulse response analysis by utilizing the coefficient value of the analyzed results from the unrestricted VAR model in Table 8. In the case of variance decomposition, Cholesky decomposition is used and the result of the variance decomposition analysis is presented in Table 9. The number of tables indicate predicted error dispersion of the variable on the left having an effect on the top variables with percentages. In the analyzed result of variance decomposition, one has to take notice of the KRX line in order to examine whether to support Hypothesis 1, Hypothesis 2, both or neither. The return rate of CME gold futures contributes to the predicted error dispersion of the market rate of return of the KRX gold market being considerably high at the 33% level, whereas the return rate of ICE dollar index futures is less than 0.4%. That is, the return rate of CME gold futures can be interpreted as having a price discovery effect that is higher than that of ICE dollar index futures on the market rate of return of the KRX gold market.

Meanwhile, when the line of the return rate of CME gold futures is seen, over 99% of it is explained by itself and the predicted error dispersion of the return rate of CME gold futures shows low importance of being explained by other variables. That is, the market rate of return of the KRX gold market is explained at a considerable level (33% level) by the return rate of CME gold futures but the opposite is not the case in this analysis. These results suggest that the price of the KRX gold market is significantly affected by the price movements of the CME gold futures market indicating the price synchronization of the two global markets.

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Table 9 Forecast Error Decomposition Analysis

This table reports unrestricted forecast error decomposition analysis results. Sample period is from the 24th March, 2014 to the 23rd August, 2016. $KRX_{co,t}$ is the market rate of return of the KRX gold market at t day, $CME_{cc,t}$ is the return rate of CME gold futures at t day, $ICE_{cc,t}$ is the return rate of ICE dollar index futures at t day. *** denotes the 1% significance level.

Division	Period	$KRX_{co,t}$	$CME_{cc,t}$	$ICE_{cc,t}$
$KRX_{co,t}$	2	67.047	32.695	0.257
	5	66.593	33.008	0.398
	10	66.592	33.010	0.398
$CME_{cc,t}$	2	0.017	99.924	0.060
	5	0.072	99.639	0.290
	10	0.072	99.639	0.290
$ICE_{cc,t}$	2	1.361	12.216	86.423
	5	1.374	12.371	86.256
	10	1.374	12.371	86.256

As shown in Table 10, the impulse responses of the market rate of return of the KRX gold market on the return rate of CME gold futures and ICE dollar index futures are analyzed. It is analyzed how $KRX_{co,}$ reacted to $KRX_{co,t}$, $CME_{cc,t}$, and $ICE_{cc,t}$ with one standard error per each given. The market rate of return of the KRX gold market at t day will react on the shock of the return rate of CME gold futures at t-1 days and ICE dollar index futures at t-1 days, taking business hours into consideration. That is, the reaction the following day needs to be examined, not the day of the impact. If the reaction of the market rate of return of the KRX gold market about the shock of the return rate of CME gold futures at t-1 days is examined, the coefficient is shown as 0.388, which is much bigger than those of the market rate of return of the KRX gold market at t-1 days (0.019) and the return rate of ICE dollar index futures at t-1 days (0.034).

This result can be interpreted as supporting Hypothesis 1 (The return rate of CME gold futures at t-1 days positively affects the market rate of return of the KRX gold market at t day) and can also support the result of the regression analysis mentioned above.

In sum, our empirical results provide that consistent relationships between ICE dollar index futures and the KRX gold market do not exist. However, the price movement of the KRX gold market reacts positively and significantly to the movement of CME gold futures while there is no significant reverse pattern.

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Table 10 Impulse Response Analysis

This table presents impulse response analysis results. Sample period is from the 24th March, 2014 to the 23rd August, 2016. $KRX_{co,t}$ is the market rate of return of the KRX gold market at t day, $CME_{cc,t}$ is the return rate of CME gold futures at t day, $ICE_{cc,t}$ is the return rate of ICE dollar index futures at t day.

	Period	$KRX_{co,t}$	$CME_{cc,t}$	$ICE_{cc,t}$
$KRX_{co,t}$	0	0.556	0.000	0.000
	1	0.019	0.388	0.034
	2	0.014	-0.051	-0.020
	3	-0.009	0.001	-0.017
$CME_{cc,t}$	0	0.007	0.977	0.000
	1	-0.011	-0.055	-0.024
	2	-0.023	-0.018	-0.047
	3	-0.004	-0.010	0.005
$ICE_{cc,t}$	0	-0.031	-0.186	0.493
	1	0.054	-0.006	-0.031
	2	-0.006	-0.023	0.012
	3	0.003	-0.002	-0.001

5. Conclusion

In this paper, we examine the opening of the KRX gold market on 24th March 2014, and also investigate the American derivatives which have an effect on the price movement of the KRX gold market. Korean gold investors found a route that they can invest in gold and save investment costs such as fees and taxes by the introduction of the KRX gold market. The implication of this paper is that the KRX gold market is developing to be the most efficient gold market among the existing gold investment methods in Korea. Moreover, despite the enormous influence of gold on other real or financial assets in Korea, the related research is scarce, especially the investigation of gold market price. In this paper, we introduce the KRX gold market and examine the causal sequence between American derivatives market and the KRX gold market, which can be a contribution to this field of research.

The futures markets could lead spot markets according to existing research on the relationship between the futures market and spot market. Moreover, when the time difference exists, the futures price at t-1 days affects the t day time point. This can be interpreted as the price discovery function. In this research, the regression analysis and VAR model are utilized and the empirical analysis is performed regarding the significance of the return rate of CME gold futures and ICE dollar index future traded at night time in Korea on the market rate of

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return of the KRX gold market the next morning. The results show that the return rate of CME gold futures at t-1 days significantly positively effects the market rate of return of the KRX gold market several hours later.

In addition, the result that the return rate of ICE dollar index futures at t-1 days has influence on the market rate of return of the KRX gold market is used for the empirical analysis in order to check whether a negative significance exists between the value of the dollar and gold price. No significance could be discovered. This result can be interpreted as supporting the existing evidence that the futures market has a price discovery effect on the spot market. However, in order to analyze empirically whether the proposition of 'the value of the dollar moves contrary to the gold price', generally mentioned in the news, etc. is supported or not, this study could not discover the result of whether or not the return rate of ICE dollar index futures at t-1 days negatively affects the significance of the market rate of return of the KRX gold market at t day. This may be due to the limit of the empirical analysis in the situation where several hours of time differences exist between the t-1 days and t day time points. Therefore, future studies need to empirical analyze not only daily data but data by minutes.

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